Tutorial 3

Randomized strategies and noisy best responses

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October 23, 2026

How this tutorial works

- Four blocks. Each one opens with a compact recap of the theory and a pointer to the lecture where it is developed, and then moves to exercises on that same material.
- The exercises use the same templates as the online quizzes on Virtuale and as the mid-term. Same mechanics, different numbers.
- Try each one first. We solve it together afterwards.
- Questions on the *content* of the course go to Prof. Bigoni. Questions on the organisation, the quizzes and the exam come to me: `marco.rosso4@unibo.it`

Coverage: Lectures 9, 10.

Plan

1. Matching pennies: why and how to randomise
2. Battle of the sexes and coordination: mixed equilibria
3. Payoff imbalances and noisy best responses
4. The playing-cards exercise of Lecture 10

Sample *oral* questions at the end.

PART 1

Matching pennies

Mixed strategies: the recipe

Lecture 9 – “Compliance and control”, “Expected payoffs”, “Summing up”

	Control	No control
Compliance	$1, -1$	$-1, 1$
No compliance	$-1, 1$	$1, -1$

No equilibrium in pure strategies: being predictable is exploited.

p = probability of Right (No control):

$$\mathbb{E}[\text{Top}] = 1(1-p) - 1 \cdot p = 1 - 2p$$

$$\mathbb{E}[\text{Bottom}] = -1(1-p) + 1 \cdot p = 2p - 1$$

$$1 - 2p = 2p - 1 \;\Rightarrow\; p = 1/2$$

Step 1. Notation: p = Pr(Column chooses Right), q = Pr(Row chooses Top).
Step 2. Expected payoff of each decision, for each player.
Step 3. Equate *Row's* expected payoffs to find p; equate *Column's* to find q.

To randomise willingly, a player must be indifferent between the decisions.

Exercise 1.1

Penalty kick – the game

Consider a soccer penalty kick situation where the kicker is equally skilful at kicking to either side, but the goalie is better diving to one side. In particular, the kicker will always score if the goalie dives away from the kick. If the goalie dives to the side of the kick, the kick is always blocked on the goalie's right but is only blocked with probability **one quarter** on the goalie's left.

Represent this as a simple game, in which the goalie earns a payoff of $+1$ for each blocked kick and -1 for each goal, and the kicker earns -1 for each blocked kick and $+1$ for each goal. (Goalie: rows; kicker: columns.)

Exercise 1.2

Penalty kick – the equilibrium

Same game as Exercise 1.1. Let p be the probability that the kicker kicks *Left* and q the probability that the goalie dives *Left*.

Determine the equilibrium probabilities used in the Nash equilibrium.

PART 2

Battle of the sexes and coordination

Battle of the sexes

Lecture 9 – “The battle of the sexes”, “An experiment”, “Expected payoffs”, “Nash equilibria”

	West	East
West	$2, 1$	$0, 0$
East	$0, 0$	$1, 2$

p = probability that Column chooses Right (East):

$$\mathbb{E}[\text{Top}] = 2\,(1-p) + 0 \cdot p = 2 - 2p$$

$$\mathbb{E}[\text{Bottom}] = 0\,(1-p) + 1 \cdot p = p$$

$$2 - 2p = p \;\Rightarrow\; p = 2/3$$

- Three Nash equilibria: two in pure strategies (upper-left and lower-right corners of the best-response graph) and one mixed, where each player chooses the preferred location with probability $2/3$.
- Experiment, 30 students, random matching: the preferred location is chosen about 67% of the time.
- With the same partner, people tend to take turns.

Exercise 2.1

A two-player game

The Row player can choose between “up” and “down”; the Column player can choose between “left” and “right”.

- If Row chooses up and Column chooses left: Row earns 5, Column earns 2.
- If Row chooses down and Column chooses right: Row earns 3, Column earns 8.
- In all other cases both players earn 0.

(a) Write the payoff table for this game.

(b) Select which category this game belongs to: a. Prisoner’s dilemma b. Matching pennies game c. Battle of the sexes d. Trust game

(c) Select all the Nash equilibria in pure strategies: a. DOWN–RIGHT b. DOWN–LEFT c. UP–LEFT d. UP–RIGHT

Exercise 2.2

Expected profits and mixed strategies

Same game as Exercise 2.1. Let p denote the probability that the Column player chooses "right", and q the probability that the Row player chooses "up".

(a) Complete the following equations for the expected profits of each player's possible choices:

$$\mathbb{E}[\pi_{\text{up}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{left}}] = \square \cdot \square + \square\,(1-\square)$$
$$\mathbb{E}[\pi_{\text{down}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{right}}] = \square \cdot \square + \square\,(1-\square)$$

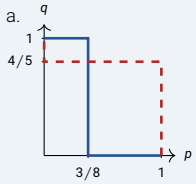
(b) The equilibrium in mixed strategy of this game prescribes that the probability that the Column player plays Right is $p = ?$, and the probability that the Row player plays Up is $q = ?$

Exercise 2.3

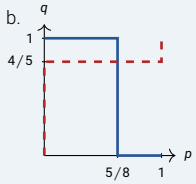
Best-response functions

Same game. Please select the graph that gives the correct representation of the best-response functions (solid: Row's best response, dashed: Column's best response).

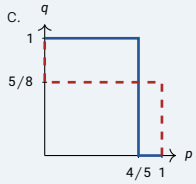
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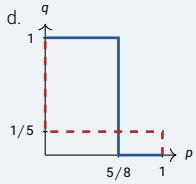
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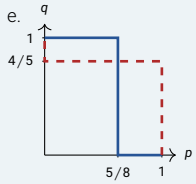
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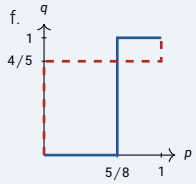
d.



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Exercise 2.4

A coordination game

Consider again the coordination game of Lecture 6.

	Left	Right
Top	3, 3	10, 0
Bottom	0, 10	20, 20

This game has two pure-strategy Nash equilibria. Does it have any equilibrium in mixed strategies? If so, find it.

PART 3

Payoff imbalances and noisy best responses

Payoff imbalances

Row’s Top/Left payoff	Row indifferent at	Column indifferent at	prediction
72 (balanced)	$p = 1/2$	$q = 1/2$	Top 50%
360	$p = 0.9$	$q = 1/2$	Top 50%
40	$p = 0.1$	$q = 1/2$	Top 50%

- In a Nash equilibrium, Row mixes to keep *Column* indifferent: Row’s own payoffs change p (Column’s mix), not q (Row’s mix).
- Data: Top rises from 36% (payoff 40) to 67% (payoff 360); Column’s Right from 24% to 74%. Behaviour reacts to a player’s **own** payoffs.

Noisy best responses and QRE

Lecture 10 – "Noisy Best Responses", "Varying the amount of noise", "Quantal Response Equilibrium"

Noisy best response: the probability of a decision rises *smoothly* with its expected payoff advantage, instead of jumping from 0 to 1.

$$\Pr(R) = \frac{\exp(\pi_R \lambda)}{\exp(\pi_R \lambda) + \exp(\pi_L \lambda)} \qquad \Pr(T) = \frac{\exp(\pi_T \lambda)}{\exp(\pi_T \lambda) + \exp(\pi_B \lambda)}$$

- Less noise: the curves get closer to the step-shaped best responses.
- **QRE**: choice probabilities match the beliefs that generate them (intersection of the noisy response curves).
- Symmetric matching pennies: the intersection stays at the centre, QRE = NE. With payoff imbalances, the QRE moves in the direction of the own-payoff effect observed in the data.

Exercise 3.1

A game

Consider a game with the following payoff matrix.

	Left	Right
Top	14, 5	5, 11
Bottom	5, 9	8, 5

Identify the pure-strategy equilibria of this game.

a. (Top, Left), with an outcome of (14, 5)
b. (Top, Right), with an outcome of (5, 11)
c. (Bottom, Left), with an outcome of (5, 9)
d. (Bottom, Right), with an outcome of (8, 5)
e. None of the above, the game does not have any equilibrium in pure strategies

Exercise 3.2

Row's expected payoffs

Same game as Exercise 3.1. Let p denote the probability that the Column player chooses "Right", and q the probability that the Row player chooses "Top".

Find the expression for the Row player's expected payoffs, one for each of Row's decisions.

Exercise 3.3

Column's expected payoffs

Same game. Let p denote the probability that the Column player chooses "Right", and q the probability that the Row player chooses "Top".

Find the expression for the Column player's expected payoffs, one for each of Column's decisions.

Exercise 3.4

The mixed equilibrium

Same game. Let p denote the probability that the Column player chooses “Right”, and q the probability that the Row player chooses “Top”.

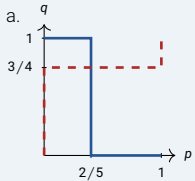
Compute the equilibrium in mixed strategies, if it exists.

Exercise 3.5

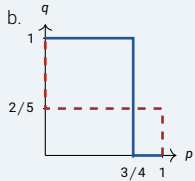
Best-response functions

Same game. Please select the graph that gives the correct representation of the best-response functions relative to the game under consideration (solid: Row, dashed: Column).

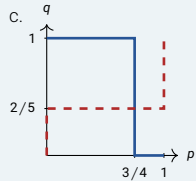
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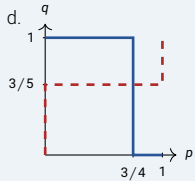
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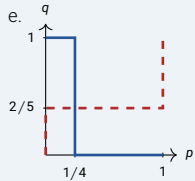
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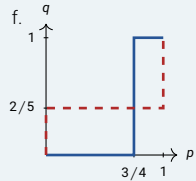
d.



e.



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Exercise 3.6

Noisy best responses

Explain what a “noisy best response” is and discuss whether the equilibrium with noisy best responses (quantal response equilibrium) would be the same as the Nash equilibrium or not.

PART 4

The playing-cards exercise

Exercise 4.1

Playing cards – Lecture 10

The row player has a red 10 (hearts or diamonds) and a black 2 (clubs or spades). The column player has a black 6 and a red 2. They have to choose which card to play, simultaneously.

- If the colours of the cards do not match (red and black), each earns nothing.
- If the colours match, earnings for each person are equal to the number on the card that the person played.

(a) Write out the payoff matrix for this game.

(b) Find all equilibria in non-random strategies.

Exercise 4.2

Playing cards – randomised strategies

Same game as Exercise 4.1.

(a) Find expressions for the Row player's expected payoffs, one for each decision, where p denotes the probability that the other player chooses their high-number card.

(b) Find expressions for the Column player's expected payoffs for each decision, black 6 or red 2, where q denotes the probability that Row plays the red 10.

(c) Find the equilibrium with randomised strategies.

Exercise 4.3

Playing cards – graph

Same game.

(a) Draw the best-reply functions of the two players, in a graph.

(b) How would the graph change if we allowed for some randomness in choices, as modelled in the Quantal Response Equilibrium?

PART 5

Oral questions

Sample oral questions I

Randomised strategies

1. Why can it be optimal to be unpredictable? Give an example of a game with no equilibrium in pure strategies.
2. Why must a player be indifferent between her decisions in order to randomise? How is the mixed equilibrium computed?
3. Describe the battle-of-the-sexes experiment of Lecture 9. How does behaviour compare with the equilibria of the game?

Noisy best responses

1. What does Nash equilibrium predict when one player's own payoff in a matching pennies game is increased? What do experiments show?
2. What is a noisy best response, and what is a quantal response equilibrium?

Sample oral questions II

Base-rate fallacy

1. To discuss the base-rate fallacy, we have considered a famous experiment in which respondents had to assess the probability that an individual (named Dick) was a Lawyer or an Engineer. Describe the experiment.
2. What is the correct reasoning according to the Bayes' rule?
3. What is instead the answer given by most of the respondents?
4. What is the base rate fallacy?